Sahaj Bhadja

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Summary

Software Engineer with 2+ years shipping full-stack **TypeScript/React** prototypes and **benchmarking** blockchain protocols against baselines, with two IEEE papers and Web3 wallet integrations.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Data Retrieval, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Shipped a multi-tenant platform on Azure SQL and Blob Storage with row-level security for 500+ users, measuring and cutting dashboard **latency** from 1.2s to 350ms through query and indexing work.
- Rebuilt async LLM job orchestration with FastAPI workers, explicit state transitions, and failure isolation, tripling **throughput** across 1K+ jobs under concurrent load and sustaining 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Shipped a full-stack **React** and Flask product end to end, from upload flows and grading history to multimodal LLM grading, serving 150+ instructors and 10K+ daily requests with 60% faster feedback.
- Built a switchable provider layer across GPT-4o, Claude, and Qwen-VL with per-call token and **cost tracking** and timeout fallback, plus concurrent batch processing of 1K+ files per run.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Optimized large-scale APIs for low-bandwidth users with pagination, field filtering, and caching, improving **response times** 30%, and automated nightly builds with 20+ test suites across three operating systems.
- Led a redesign of ETL workflows on AWS Glue and S3, boosting **pipeline efficiency** 68% and cutting storage costs 15%, while collaborating with Microsoft and GitHub stakeholders in Agile sprints.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship *India*

- Designed a **quantum-resistant blockchain** pairing lattice-based FALCON signatures with node-monitored delegated proof-of-stake consensus and Borda Count witness selection, cutting centralized-control and quantum-risk exposure 25% in benchmark simulations.
- **Benchmarked** the protocol against an ECDSA baseline across 100–1,000 transaction workloads, sustaining 957.8 TPS at 224ms versus 968.3 TPS and 211ms, and presented results at IISc Bengaluru.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Integrated **Web3** wallet-signature authentication with web3.js and MetaMask for passwordless login, eliminating password storage for wallet users while handling wallet connectivity issues on React Native mobile clients.
- Built Python Flask REST APIs with Firebase **real-time** sync, transactional locking, and idempotent payment webhooks, sustaining sub-200ms latency and 40% faster booking confirmations across 50K+ booking records.

Skills

Languages: TypeScript, JavaScript, Python, Solidity, Java, C++, SQL, Git, Linux
Full Stack: React, Next.js, Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, PostgreSQL, MongoDB
Blockchain: Consensus Algorithms, Lattice-Based Cryptography, Smart Contracts, web3.js, MetaMask, Truffle, Ganache
Performance: Benchmarking, Latency Analysis, Throughput Analysis, Distributed Systems, Query Optimization, Caching, Concurrency
Cloud & Practices: Firebase, AWS, Docker, Kubernetes, CI/CD, Rapid Prototyping, Claude Code, Technical Writing

Projects

P2P Gossip Benchmark Harness | *TypeScript, Node.js, js-libp2p, Docker, React, Python*

- Designed a benchmark harness that spins up Dockerized js-libp2p nodes on configurable topologies, compares **GossipSub** against alternative propagation strategies on p50/p99 **latency** and delivery rate, and charts results in React.

FinNexus Live | *Next.js, TypeScript, WebSockets, MongoDB, Kubernetes*

- Shipped a real-time **TypeScript** and Next.js fintech app with WebSocket updates, Plaid bank linking, and a token-based firewall, deployed on Kubernetes for 50K+ users across 1M+ transactions.

Comprehensive Analysis of Blockchain for User Authentication Paper | *Smart Contracts, DLT, Self-Sovereign Identity*

- Co-authored an IEEE paper analyzing **blockchain** authentication with smart contracts, distributed ledgers, and self-sovereign identity for users and IoT devices, including replay, man-in-the-middle, and phishing threat models.